

Sienna Giuseppi Q5 Design Summary

This project challenged me to rethink how a product I use every day could become more energy efficient through mechanical design. Revisiting the laptop allowed me to build upon research I began in high school using the mechanical engineering knowledge I have gained over the past two years.

I began by analyzing the complete power pathway of my Microsoft Surface Laptop Studio 2 during a typical web browsing session. Using battery discharge measurements, Windows battery reports, and first-principles energy calculations, I established a quantitative baseline for the laptop's average power consumption before even considering how to improve it.

My initial thought was that I needed to invent something completely new. I spent hours digging through research papers on everything from piezoelectric and triboelectric energy harvesting to spacecraft thermal management, Formula 1 cooling systems, refrigeration, and battery technology. I kept asking myself, *what cutting-edge technology could I borrow and adapt into a laptop?*

From there, I explored several redesign concepts, beginning by revisiting my high school research on piezoelectric computer keys before investigating thermoelectric energy recovery. While both concepts demonstrated that energy could theoretically be harvested from the user or from waste heat, the amount of recoverable energy was simply too small to meaningfully improve the efficiency of a laptop. Every new idea added complexity while recovering very little power.

That realization completely changed my approach. Instead of asking how I could generate more energy, I started asking how I could stop wasting so much of it in the first place.

It took a surprisingly long time before I finally stepped back and looked at what was already there. I didn't want to completely reinvent the laptop. Instead, I wanted to understand the technologies that already exist and see if they could simply be organized more effectively. While researching thermal management, I found countless Reddit discussions and YouTube videos of enthusiasts modifying their own laptops. One modification kept appearing over and over again: adding thermal pads that transfer processor heat into the aluminum chassis. Users consistently reported lower processor temperatures and better sustained performance because the chassis acted as a larger heat spreader, but the laptop also became noticeably hotter to touch.

That observation became the turning point of the project. Instead of thinking purely as an engineer, I started thinking like a customer. I realized I almost always touch the palm rests and bottom panel, but rarely the rear hinge or side rails. If those regions are already exposed aluminum and are rarely touched, why shouldn't they become the primary locations for passive heat rejection? Instead of treating the entire chassis equally, the laptop could intelligently decide *where* heat should be released.

This led to my final concept: Selective **Thermal Architecture**. Rather than replacing the existing cooling system, the design introduces a parallel passive cooling pathway using laser-patterned pyrolytic graphite sheets to route heat toward designated radiator regions along the rear structural beam and side rails. Internal thermal necks increase thermal resistance between these radiator regions and high-contact areas such as the palm rests, while CNC-machined microchannels and high-emissivity coatings improve passive heat rejection without compromising the clean unibody design of the laptop.

One aspect of this concept that I appreciate most is its practicality. Rather than relying on emerging technologies that I do not yet fully understand, such as advanced phase-change cooling systems or entirely new materials, I built the design around technologies that already exist in modern laptops and simply reorganized them into a different thermal architecture. Because the proposal functions as a parallel passive cooling system, I believe it could be adapted to many existing laptops without requiring an entirely new computer architecture.

The most valuable lesson I learned from this challenge is that good engineering is not always about inventing completely new technology, but about understanding existing systems well enough to reorganize them into something more effective. I began this project believing the answer to sustainability was generating additional clean energy. I finished it realizing that the greatest opportunity often lies in designing products that simply require less energy to accomplish the same task. That shift in perspective fundamentally changed how I think about engineering design, and it will influence the products I design moving forward.

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